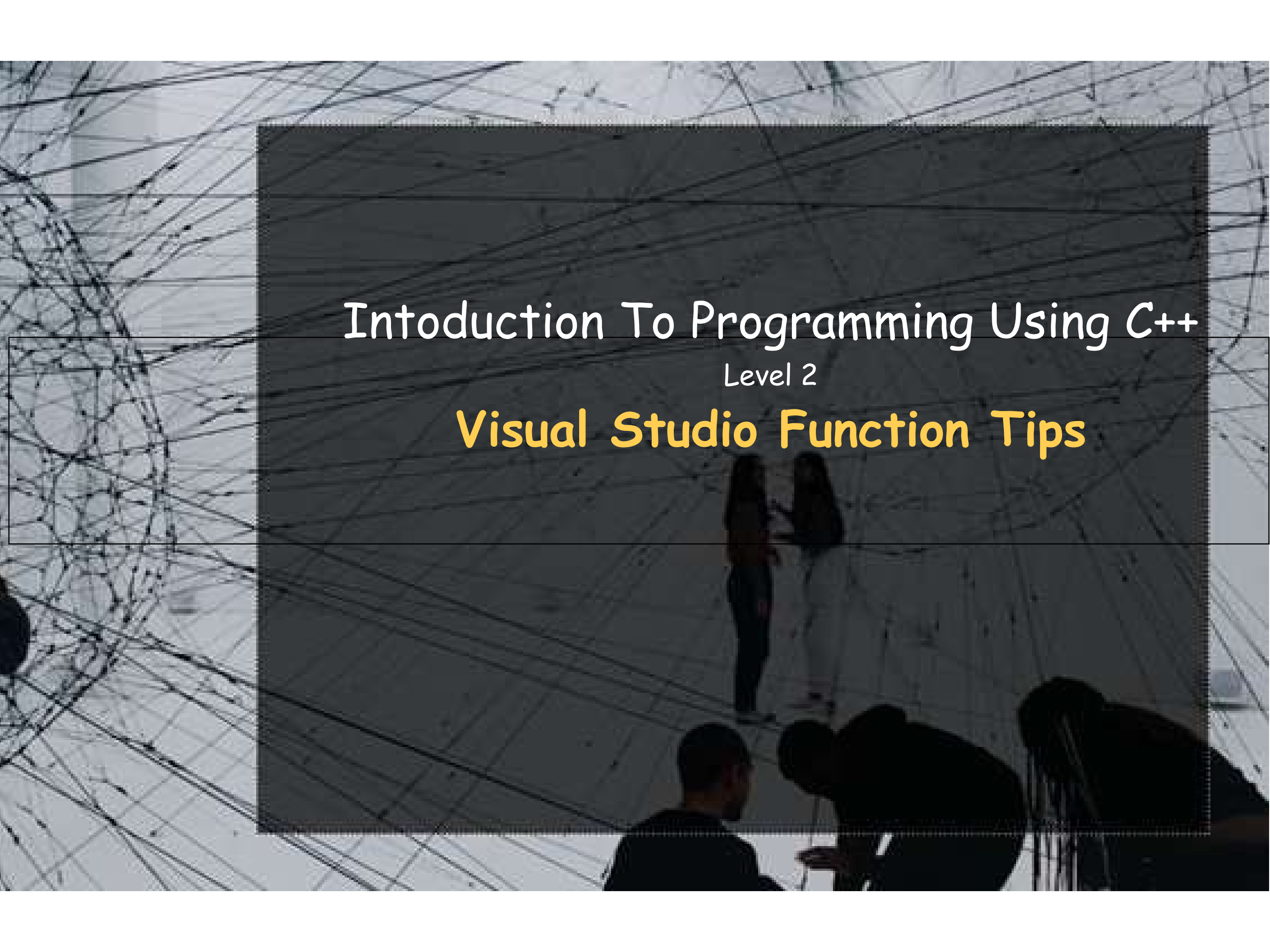




Introduction To Programming Using C++ Level 2

Visual Studio Function Tips



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Level 2

Visual Studio Function Tips

Definition

◆ Visual Studio provides several useful tools that make working with Functions easier and help reduce complexity when viewing and navigating code.

Explanation

 Visual Studio allows us to Collapse and Expand Functions to hide or show their contents.

 It also provides tools to quickly find where a Function is declared or defined, see where it is used, understand its Call Hierarchy, and Rename it safely.

Procedures

1. To Collapse or Expand a Function, click the small arrow next to the Function or use the corresponding Visual Studio shortcut.
2. To find the Function Definition, right-click the Function and select `Go To Definition`.
3. To find the Function Declaration, right-click the Function and select `Go To Declaration`.
4. To see which Functions call the selected Function, or which Functions it calls, select `View Call Hierarchy`.
5. To find every place where the Function is used, select `Find All References`.
6. To view the Function Definition without leaving the current location, select `Peek Definition`.
7. To safely change the Function name, right-click it and select `Rename`.

💡 Example

```
void Function2();  
void Function4()  
{ cout << "Hi I'm function4 " << endl; }  
void Function3()  
{ Function4(); }  
void Function2()  
{ Function3(); }  
void Function1()  
{  
    Function2();  
    Function4();  
}  
int main()  
{  
    Function1();  
  
    return 0;  
}
```

➡ Result

◆ We can use Visual Studio tools to quickly navigate and understand the relationships between the Functions.

Note

Collapse / Expand

-  Hides or shows the contents of a Function.

-  Useful when working with large Functions or large source files.

Go To Definition

-  Opens the complete Function Definition.

Go To Declaration

-  Opens the Function Declaration.

View Call Hierarchy

-  Shows which Functions call the selected Function and which Functions are called from it.

Find All References

-  Shows all places where the selected Function is referenced or used in the program.

Peek Definition

-  Shows the Function Definition in a small preview window without navigating away from the current code.

Rename

-  Allows you to rename a Function and update its references safely.

⊘ Warning

- ◆ When renaming a Function, it is better to use Visual Studio's `Rename` feature instead of manually changing the name in each location.
- ◆ This reduces the risk of missing one of the Function's references.



Comparison

▨ Navigation Tools

- ✓ `Go To Definition` → Opens the complete Definition.
- ✓ `Go To Declaration` → Opens the Declaration.
- ✓ `Peek Definition` → Shows the Definition without leaving the current location.

▨ Function Analysis Tools

- ✓ `View Call Hierarchy` → Shows Function calling relationships.
- ✓ `Find All References` → Shows where the Function is used.
- ✓ `Rename` → Changes the Function name and updates its references.

✦ Summary

- ✓ Visual Studio provides useful tools for working with Functions.
- ✓ `Collapse / Expand` helps organize the code view.
- ✓ `Go To Definition / Declaration` helps locate Function information.
- ✓ `View Call Hierarchy` helps understand Function Calls.
- ✓ `Find All References` shows where a Function is used.
- ✓ `Peek Definition` shows a Function Definition without leaving the current location.
- ✓ `Rename` helps safely change a Function's name.

Thank you 😊